Beyond intelligibility — the performance of text-to-speech synthesizers

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Performance assessment tests using opinion scales and subjective measures have long been used as the basis for rating the speech quality of telephony transmission systems. Recently these methods have been adapted to speech synthesiser evaluation and used to develop a standard test methodology which has been applied to most of the text-to-speech systems which are currently available in the market. In the course of this work a new reference has been defined. The procedure draws heavily upon current CCITT (ITU) Recommendations relating to subjective evaluation and enables comparisons to be made between results obtained in different experiments. The results indicate that comparisons made using the new reference system are consistent across subject groups and are tolerant to different types of speech material and opinion scoring methods.

1. Introduction

Although text-to-speech synthesis offers the most flexible way of generating spoken messages, it has yet to find widespread use within telephony systems. There are several reasons for this, but the main one is that the perceived quality of synthetic speech has always lagged behind that of recorded natural speech. Only in very specialized niche applications has the advantage of flexibility been sufficient to outweigh the reduced performance.

However, text to speech (TTS) remains the only viable solution for all those applications where unconstrained text must be turned into speech, and the demand is increasing as more text becomes available on-line. The current growth of spoken e-mail provides a particularly important example (see Chapter 5) and there has been an increase in the number and variety of techniques now used (see Chapters 6 and 7). Given this interest there is a need to evaluate the performance of TTS systems, both to compare alternative methods and to determine how systems complement one another.

This chapter begins by reviewing the traditional methods used for telephony transmission performance and describes how they have been adapted for evaluating the performance of synthetic speech. It focuses on subjective methods and explains why tests based upon opinion scores have come to supersede other types of test for ‘toll-quality’ speech. It describes the reference system which has been developed as a standard for TTS evaluation and shows how it has been validated as a reliable technique. Finally the results of tests on a range of contemporary text-to-speech systems are presented.

2. Subjective testing methods used in telephony

Methods based upon subjective experiments using opinion scales and reference systems are now so well established that they are routinely used for planning and improving the speech quality of telephone transmission systems [1].

The most general type of test is the ‘conversation’ test, in which pairs of subjects converse over various transmission links. Conversation tests take into account sidetone, overall loss, delay and echo as well as the quality of received speech, and so include all the factors which are likely to impact upon the perceived quality of a normal telephone conversation. Chapter 14 discusses some of the related perceptual issues, and Chapter 4 shows how such tests are used for transmission performance evaluation.

However, not all speech is conversational. Recorded announcement services have been available for over 50 years (the Speaking Clock was introduced in 1936) and although glass discs with analogue signals have been replaced by plastic discs with digital signals, many of the latest recorded information services still reproduce speech by concatenating prerecorded speech fragments.

In such cases listening tests provide an alternative to conversation tests, and over the years a number of procedures and recommendations [2] have been developed. It is these which have formed the basis for the methods presented here.

3. Why should transmission methods be used for evaluating synthetic speech?

As the methods used for telephony speech quality evaluation have traditionally been so focused on coded speech, it may be argued that, as synthetic speech is not the same as coded speech, different methods should be used to evaluate it.

For some market segments (e.g. talking toys) this may well be the case, but for the vast majority of telephony-based applications, text-to-speech systems will inevitably have to compete 'head to head' with alternative technologies. These will include 'live' human speech, pre-recorded messages and concatenated speech. Such an approach is already successfully deployed in BT's current directory enquiries systems where a concatenated digit string is appended to a pre-recorded message. Some call-centres already intersperse 'live' speech with pre-recorded and synthetic speech, and with current advances, which allow the synthesizer to have the same 'voice' as the recorded speaker, these hybrid approaches will become increasingly common.

For telephony applications, therefore, the assessment method must be capable of comparing and contrasting all types of computer-mediated speech. This requirement has been a major driver for the method developed here.

4. Background to speech-quality evaluation

There are two main groups of people for whom telephony-based speech-quality evaluation is useful:

- the developers of the technology;
- the users of the technology.

Their requirements are different. By and large the developers want to know if a change to an algorithm or technique has provided a real improvement. Ideally the test should be diagnostic — identifying what it is about the change that has brought about the desired effect.

Users, on the other hand, normally want to compare different commercial offerings of the same technology — often from different suppliers. Sometimes they may even want to differentiate between entirely different technologies. Ultimately, they want standards which can be applied to as broad a range of candidate technologies as possible and encourage the growth of a commodity market.

Over the years a wide range of methods have been developed to serve the needs of both groups. All of these test methods tend to fall into one of a number of broad categories, each of which is sensitive to speech qualities ranging from the barely intelligible to high fidelity.

Some of the major 'quality bands' are shown in Fig 1. The application areas served by the particular range are shown at the bottom of the figure and the double arrowhead lines highlight the ranges over which different test methodologies provide discriminative responses. The categories are defined in the following sections.

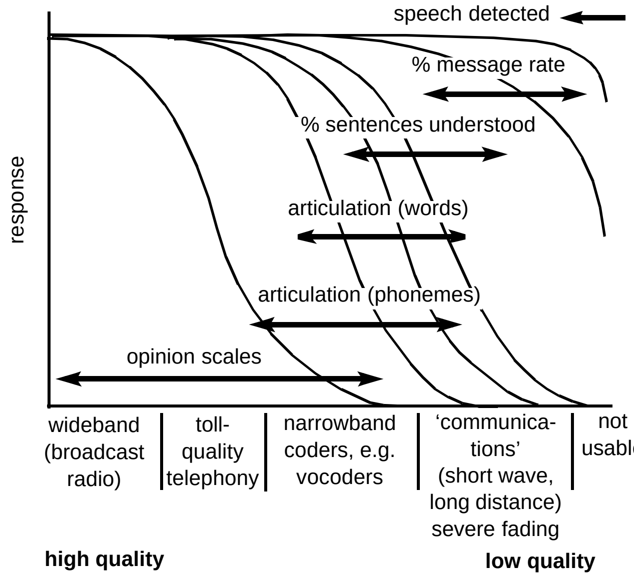

Fig 1 Relationships between quality bands, test methods and typical applications.

4.1 *Tests based upon detection of thresholds*

Apart from total absence, the lowest quality of speech is that which is just detectable. At this level the speech cannot be understood — but a listener will be able to identify it as speech. Such speech need not just be very quiet (audible), it may be buried in noise or grossly distorted. This point is shown at the top right of Fig 1. In principle it may be determined by asking a number of listeners to listen to the signals in question and register when they can detect speech. As the quality improves, other thresholds such as those determined by usability, intelligibility and comfort are reached. However, once any of these thresholds has been passed, the test becomes insensitive to further change. For example, once a system has passed the usability threshold, it is no longer useful to base quality on a usability index. The onset of a threshold effect can be quite sharp — as indicated by the 'knee' in the detectability characteristic — but may be blurred if different types of test material are employed. It may also vary considerably between individual listeners.

The range of sensitivity may sometimes be extended by exploiting masking phenomena where subjects are required to detect signals in the presence of added noise or other structured signals. In general, their role is limited to those cases where similar impairments are likely to appear in real applications. As they do not provide a smooth gradation of quality indices, they are not really suitable for overall quality evaluation [1].

4.2 Methods based upon measuring the performance of specified tasks

Just above the threshold of audibility, lies a range where speech is detectable, but the quality is such that there is information loss and frequent repetition or rephrasing of messages is required for meaning to be understood. 'Message rate' metrics such as transaction time, speed of response or job rate are sometimes considered appropriate here.

Apart from the fact that such tests span a range which borders upon being usable, there is also a question as to what is actually being measured, as task responses are confounded by factors such as the subjects' intelligence, stamina, motivation and dexterity. These are 'nuisance effects' in statistical terminology. It is possible to eliminate their perturbing effects by careful statistical design, but only at the cost of a large increase in the amount of work required.

One method occasionally used to try to extend such tests to a higher range of qualities involves the use of a simultaneous secondary task which increases the amount of work (cognitive load) required of the subjects. The secondary task is chosen to be simple and directly measurable, e.g. the subject may be required to repeatedly type the alphabet into a keyboard while simultaneously responding to separate spoken instructions from the synthesizer under test. The quality of the speech synthesizer is then assumed to bear an inverse relationship to the rate at which the secondary task is performed. Apart from the uncertainties highlighted above, this method makes the additional, rather dubious, assumption that subjects are limited by a 'finite cognitive capacity.'

There are some reports of such methods being used for text-to-speech evaluation [3], but most modern synthesizers have a quality above that measurable by such tests.

4.3 Articulation tests

In message rate tests subjects hear complete messages and so are often able to use context to 'fill in' those parts which are misheard. By shortening the messages to tokens consisting of single words, syllables or even phonemes, they can no longer do this.

Tests based on such tokens are called 'Articulation Tests', and like message rate tests rely upon there being information loss, the extent of which can be directly measured by asking listeners to label what they hear. Probably more than any other, this is the type of test which has been most widely used among text-to-speech researchers [4]. Typical of articulation tests are:

- intelligibility tests;
- rhyming/analogy tests.

The sounds used may be familiar, distinct words or syllables, in which case one range of qualities is covered. More often they are combinations of phonemes into nonsense syllables known as logatoms — this technique extends slightly upwards the range over which discriminating responses can be obtained. Usually the quality of the received speech is scored according to how well the subjects identify the correct token as part of a 'closed set' of candidates, although in rhyme tests they need only indicate a close match. Such tests have been used in synthesizer development where it is necessary to determine which algorithmic variant produces sounds closest to a 'target' sound [5].

Although the range of quality is higher than that used for message rate tests, by definition articulation tests are only valid where the speech quality is so bad that intelligibility is compromised. As such their role has been confined to evaluating very low bit rate vocoded signals, narrow-channel systems occasionally used in military or police applications, or very poor 'public-address' systems.

Once a speech signal has breached the 'intelligibility threshold' articulation tests lose their ability to discriminate. As most of today's text-to-speech systems are above this level, albeit still demanding effort on behalf the listener, methods based upon articulation scores can no longer be considered suitable for text-to-speech evaluation.

4.4 Tests based upon equivalence

Tests based upon equivalence form a very powerful and wide-ranging set of tests which can be used for almost every quality band. One of the simplest is based upon pair comparison, where subjects are required to decide which of two signals is preferred. Such tests are widely used when a new or alternative design (e.g. of microphone) or process (e.g. companding law) is being considered as a replacement for an established system.

The subjects' preference may be open (Which do you prefer?), or may be focused on a particular attribute — loudness, distortion, pleasantness, etc. In one variant the subject may adjust one source (the reference) until both are judged to be the same. Impairments being compared need not be identical — subjects may be asked to select which of two entirely different distortions is more pleasant or more comfortable. Such methods have been widely used for setting standards in telecommunications, for example, to compare the quality of a digital coding scheme to that of frequency division multiplexing.

In a single pair comparison, such tests only determine the ranking of systems, but tests based upon equivalence become especially powerful and general when they are used with a reference system. By finding the equivalent setting of a reference system it becomes possible to rate an unknown system against a fixed and well-defined standard.

Equivalence tests and preference tests (see section 8.4.5) are closely linked, the difference being the extent of generality. In equivalence tests, subjects are required to judge that two systems are similar 'head to head'. In opinion-based tests, comparisons of more than two systems can be made.

4.5 Tests based upon opinions

All of the above methods rely upon subjective responses, but, in most cases, the role of the subject is reduced to that of a 'detector'. However, the fact that humans remain sensitive to very small amounts of distortion even when quality is already high suggests that they could also be used directly as measuring instruments. The only problem lies in devising methods to exploit this sensitivity. The solution is simple — ask subjects directly for their opinion about the perceived quality. At first sight such an approach appears 'non-scientific' and surrounded with difficulties. It may be argued that every listener will have a different set of values and expectations, likely to cause their responses to fluctuate wildly and make analysis impossible. Moreover, their opinions will be influenced by all sorts of other factors — the 'accent' of the voice, their understanding of what is being spoken, even how comfortable they feel about giving their opinions.

However, it is precisely because people's opinions are so sensitive, not just to the signal being heard, but also to norms and expectations, that opinion tests form the basis of all modern speech-quality assessment methods.

Provided that the experiment is based upon a good statistical experimental design and the appropriate analysis methods are used [6], then not only can significant results be obtained from a small number of representative subjects, but the results will be a good predictor of how the total population will react. As a consequence, not only does the method directly address the quality dimensions of interest, but it also directly aligns the results with customer expectations.

By judicious choice of the opinion scale used, tests can be designed to cover any desired range of qualities. For telephony-quality speech two opinion scales now tend to predominate and are established as ITU standards:

- the quality scale:

  excellent,

  good,

  fair,

  poor,

  bad,

  is a particularly general scale used for both conversation and listening type tests;

- the listening effort scale **(effort required to understand the meaning of sentences)**:

  complete relaxation possible — no effort required,

  attention necessary — no appreciable effort required,

  moderate effort required,

  considerable effort required,

  no meaning understood with any feasible effort.

There is a distinction to be drawn between the responses obtained from each. The listening effort scale applies to speech, whereas the quality scale may be applied to the communication link.

One other common scale is the loudness scale, which, although not a quality scale, may be used for determining the optimum level or range of loudness for comfortable listening. How the same method is used for evaluating very high quality speech is described in the ITU Recommendation [7].

Such methods can be standardized when coupling them to a separate reference system. Then, not only can results obtained at different times or places be more directly and reliably compared, but they may be used for all languages. Section 8.5 describes these in more detail.

4.6 Test based upon acceptability

This section would not be complete without mentioning 'acceptability'. All of the above tests are designed to allow comparisons to be drawn between different technologies. The assumption is made that if one technique is perceived to be of higher quality than the others, then that will also be the most acceptable for any application. That acceptability is relative can be illustrated by the plight of King Richard III:

'A horse, a horse: my Kingdom for a horse.'

It seems unlikely that in more normal times King Richard III would have considered his kingdom to be an acceptable exchange for a horse — yet clearly this had become the case.

This example serves to illustrate that any notion of acceptability is determined by the application, the alternatives on offer, user motivation (including cost), and how pressing the need. It is not an intrinsic property of a technology — nor a horse — and cannot be determined by direct measurement.

The only way in which acceptability can be 'fixed' is by comparison with an arbitrary standard — usually defined in operational terms. For example, in telephony, the level of acceptability was once defined as 'the quality at which 5% of subscribers on limiting connections (lines with more than 30 dB of loss) complained.'

Once established, the minimum specification of a system which meets this standard can be defined. No such operational standard has yet been set for text-to-speech systems — but, as will be shown later, the use of a reference system opens the way for such a standard to be defined and specified.

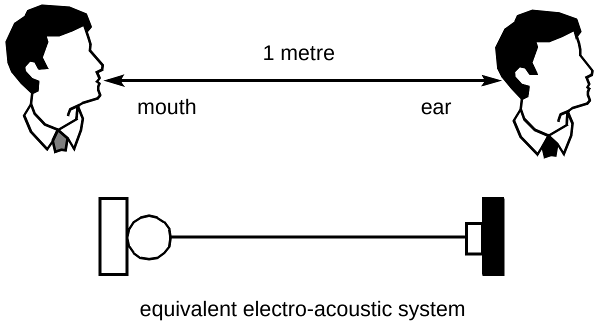

Fig 2 Electro-acoustic equivalent of one-metre air path.

5. Reference systems

The first reference system designed for rating telephone speech quality was based upon the notion of the 'one-metre airpath' — see Fig 2.

'Standard' speech was defined as being that which was received at the ear of a listener one metre from the lips of a speaker. This provided a 'gold' standard which had the merit of being simple, was linked to other physical standards, and was independent of language, talkers, listeners and the technologies to be tested.

However, the standard only defined a single quality. In order that it could be used to rate other speech qualities, some means had to be found whereby it could be controllably impaired. The simplest way to do this would have been to progressively increase the air-gap in standard steps — hence increasing the loss. Speech could then be rated as equivalent to that heard over 2 m, 3 m, 50 m, etc. However, this was not considered realistic and the standard eventually chosen was based upon an electro-acoustical equivalent of the one-metre air path. It was was known as SFERT and was defined in 1928. A number of other models followed, e.g. ARAEN, SRAEN and NOSFER which embodied improved technical implementations [8] (see Fig 3).

Today the standard is defined in terms of the measured electrical and acoustic properties of an intermediate reference system (IRS). The present IRS is defined by CCITT (Blue Book) Recommendation P.48 (Fig 4).

5.1 The impairment principle

The impairment principle allows a real system to be rated against the standard. The principle is illustrated in Fig 5.

Fig 3 AREAN (1948) being used to rate a telephone connection. The specially trained speaker can be seen to be speaking into the handset being rated and into the AREAN 'ball and biscuit' microphone. The instrument on the right hand side of the picture is the meter of the Speech Voltmeter No 3 which allows the speaker to monitor, and keep constant, her speech level.

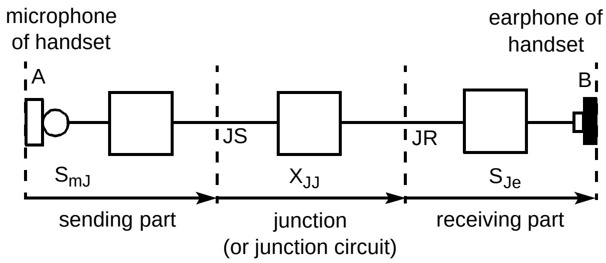

Fig 4 Diagram of IRS.

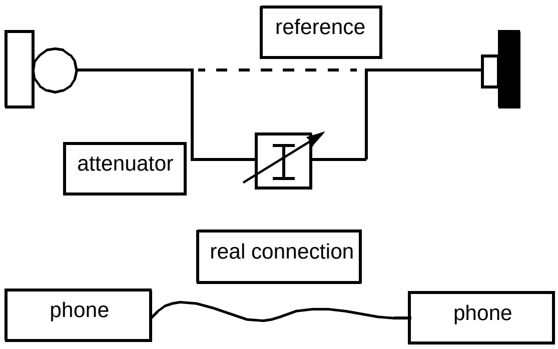

Fig 5 The impairment principle.

The idea is that any 'real' telephone connection has a performance which is 'equivalent' to the reference system impaired by a certain amount. For example, a very long-distance connection might be found to be equivalent to the reference system in combination with a 20-dB loss in the transmission section. By an obvious extension of the same principle, a digital circuit might be found to be equivalent to the reference system with noise injected at 30 dB below the signal level.

The determination of such equivalents can only be obtained subjectively. To undertake such tests it is necessary to determine the reference impairment, the speech material, and the criteria upon which subjects base their decisions.

5.2 *Choice of impairments*

For early telephony systems the dominant problem in the 'real' network was loss, and a simple attenuator provided an appropriate method for providing a realistic and controllable impairment. For other networks, however, alternatives had to be found. Added noise was used for some frequency division multiplex circuits, and injected 'babble' (noise shaped with a speech spectrum) was used to provide a reference for crosstalk. Probably the best known and most widely used of telephony reference standards is the Modulated Noise Reference Unit, which provides an impairment suitable for rating many digital networks [9].

5.3 *Choice of material*

A choice must be made concerning the speech material used in tests. If the application is known, then the best practice is to use speech that is representative of that application. For telephony systems (and for most TTS applications), this is almost a completely open set, but clearly excludes the use of 'nonsense' material, foreign-language material, and phrases which contain specialist terms or jargon.

The traditional material employed at BT has been based on no mental effort (NME) sentences. These are simple sentences between 5 and 10 words in length selected at random from contemporary novels. They are examined, and any which appear to be complex, ambiguous or contain grammatical constructs, with which some listeners may not be familiar, are removed. Typical of such sentences are: 'The sun shone brightly in a blue sky', and 'The table was covered with a blue cloth'. The meaning of the sentences is deliberately made as simple as possible to try to ensure that the opinions of the subjects are mainly influenced by the signal quality of the material heard — not by its semantic aberrations. [1]

[1] We try **not** to draw the subject's conscious attention to the signal quality any more than is necessary; what we want them to think about, at least in listening effort tests, is how easy or difficult they found it to understand.

6. The BT reference system

Although general guidelines for all of the above are provided by CCITT [10], specific choices still had to be made to provide a system which was suitable for TTS evaluation. Those choices were determined as follows.

6.1 Choice of impairment

For a rating method to be successful, the impaired reference signal must sound similar to the systems under test and to find such an impairment a number of candidates were investigated. First among these was the modulated noise reference unit [10], already used for rating digitally encoded speech in telephony systems. As a well-established standard it would have been ideal if this could have been used without modification. However, it was immediately apparent that the MNRU impairments did not bear any similarity to those produced by any synthesizer; nor did any of the other 'classical' impairment methods — loss, injected noise, babble — transform 'real' speech into a form which could conceivably be considered similar to synthetic speech.

A number of 'novel' candidates were considered next. These included various modulation schemes, e.g. ring modulators to give 'Dalek'-like properties, and several low bit rate speech-coding schemes. Only two produced speech which was plausibly 'synthesizer-like'. These were linear predictive coding [11] and what came to be known as time and frequency warping (TFW).

LPC provided speech which, not unsurprisingly given that a number of synthesizers are LPC based, sounded closest to some synthetic speech systems. However, there was no known way by which an LPC system could be made continuously and monotonically degradable with the required precision. The use of an impairment so closely aligned with one particular method of synthesis was also undesirable.

TFW speech, however, could be varied over a wide range of qualities, and the parameters could be set to simulate some of the grosser surface effects, such as the durational and spectral aberrations present in synthetic speech.

6.2 Description of TFW

The time and frequency warping method is nothing more than carefully controlled 'wow' and 'flutter' caused by speeding up and slowing down the speech signal. In the original model the speech signal was recorded at a constant sampling frequency and then played back at a variable rate as shown in Fig 6.

The process has three variable parameters — modulation amplitude, modulation period and modulation waveform. After some initial trials a sinusoidal modulation waveform with a period of 150 ms was chosen so that the

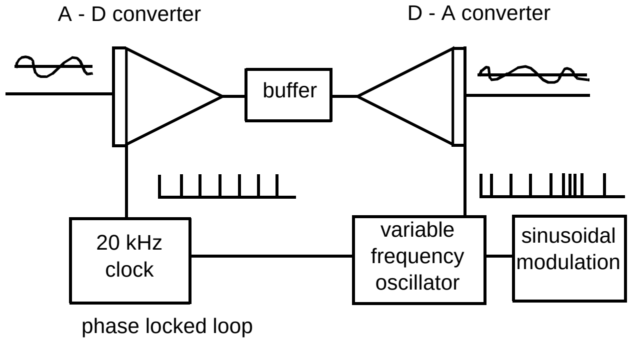

Fig 6 Basis of the TFW impairment process.

only variable parameter was the amplitude of the modulation about the original sampling rate. The specific algorithm is as follows.

Sampling of original speech signal

Let S_i be the ith sample taken at time T_i when sampled at a constant sampling rate F_s (Hz), where $i = 0, 1, 2,..., k$ and k is the total number of samples.

Hence $T_i = \frac{i}{F_s}$

TFW playback

Sample S_i is then replayed at time T'_i, where:

$$T'_i = \frac{1}{F_s + A_w \sin\left[\frac{2\pi i F_w}{F_s}\right]}$$

and A_w and F_w are the amplitude and frequency of the required warping $i = 0, 1, 2,..., k$.

6.3 Implementation and refinement

In the original design, special hardware was required to synchronize the clocks, but the present system is entirely software based. Speech sampled at a fixed rate of 16 kHz and 16 bits resolution is processed 'off-line'. The software resamples the processed file using appropriate interpolation to calculate the magnitude of the 'warped' samples.

The software TFW algorithm has three main parameters — the amplitude, the rate and the shape (sine, inverse sine, triangular, etc) of the deviation. After having conducted a number of pilot tests, it was decided to retain the original warping period of 150 ms and to sinusoidally swing the frequency to a maximum of 3.2 kHz either side of the 16 kHz, at which level of distortion the processed speech file is almost unintelligible to the naive listener.

6.4 Calibration and validation

Calibration and validation of the reference system was undertaken in a number of experiments to confirm that the system spanned the necessary ranges and to check for monotonicity. Figure 7 shows how the system performs using listening effort (LE) as a parameter and shows that the design is both sensitive to listening level and exercises the full range of the response scales.

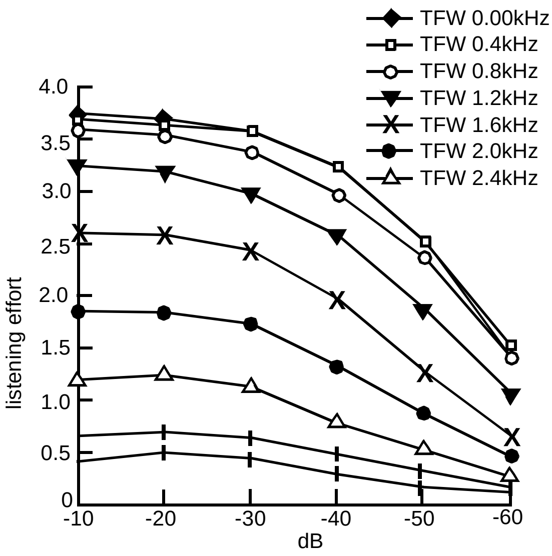

Fig 7 Listening effort versus level (TFW as paramter).

7. Application experience

This software version TFW reference has now been used for several years to monitor and track performa nce of BT's own synthesizer [1], and to benchmark a number of commercially available systems with British English or American English output. So far five test series have been undertaken with three synthesizers (or variants of synthesizer) normally included in each, and in this section the results from several of these test series are presented and explained.

7.1 *Principal procedures adopted in the tests*

In each test, subjects are drawn from a population of untrained naive listeners. The main factors (recorded reference speech, synthetic speech and choice of speech material) are arranged and balanced using standard statistical methods of experimental design and randomized orthogonal squares to minimize bias.

The tests are carried out using the procedures outlined in CCITT P.48 [12] in a sound-insulated room with 50 dBA Hoth noise.The material is presented over a range of fixed listening level loudnesses, from a loudness above that which is comfortable, to a level below that which would be encountered in modern telephone systems.

Each listener hears a series of samples of speech (synthetic and reference) in some predetermined random order. After each sample, the subject indicates his or her opinion of the effort required to understand the meaning of the speech. Subsequent treatment is based upon allocating numerical scores to these responses. A 'repeated measures analysis of variance' [6] is performed to identify which factors in the test are significant.

7.2 *Main results of experiments*

To illustrate the general trends, the results from a total of ten listening experiments are presented (see Table 1)

Table 1 Summary of experiments.

Experiment	Synthesisers	Opinion scale used	Speech material	Subjective results shown in
A	Laureate + A&B	Listening effort	NME Sentences	Fig 8
B	Laureate + A&B	Listening effort	Paragraphs	Fig 9
C	Laureate + A&B	Quality scale	NME Sentences	Fig 10
D	Laureate + A&B	Quality scale	Paragraphs	Fig 11
E	Laureate + C&D	Listening effort	NME Sentences	Fig 12
F	Laureate + C&D	Listening effort	Paragraphs	Fig 13
G	Laureate + C&D	Quality scale	NME Sentences	Fig 14
H	Laureate + C&D	Quality scale	Paragraphs	Fig 15
I	Laureate + E&F	Listening effort	NME Sentences	Fig 16
J	Laureate + E&F	Listening effort	Paragraphs	Fig 17

The first four experiments (A-D) were carried out on three synthesizers. One of these was Laureate — BT's in-house synthesizer — and the other two were commercially available systems running on PCs. The results are shown in graphical form in Figs 8-11.

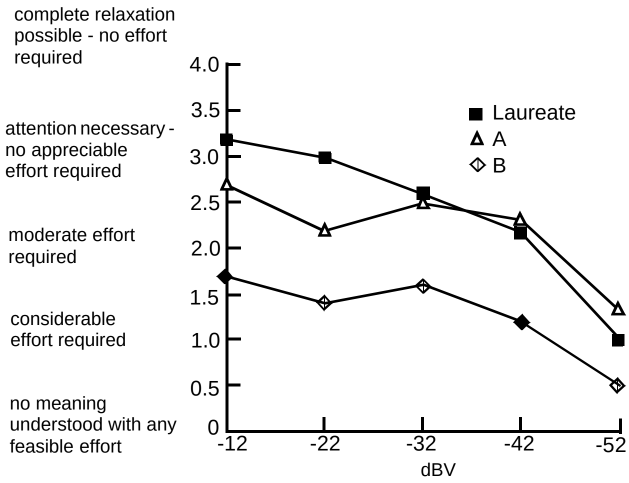

Fig 8 Experiment A — listening effort, NME sentences.

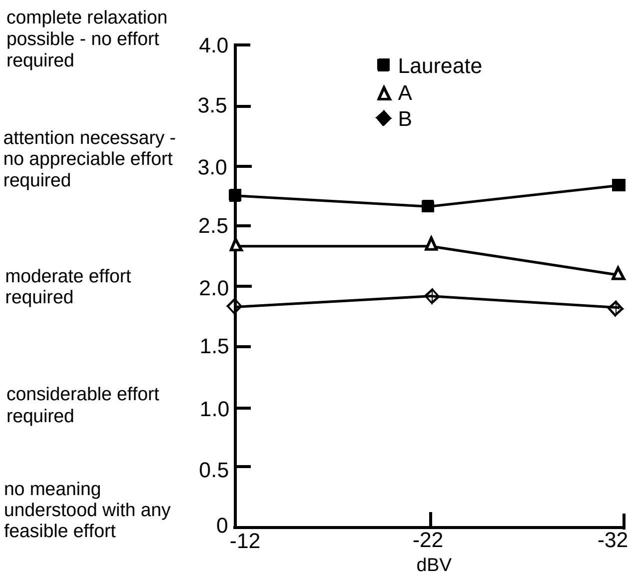

Fig 9 Experiment B — listening effort, paragraphs.

The second set of experiments were the same as the first, except that two different commercial synthesizers were used, and results are shown in Fig 12-15.

The third set of experiments used another pair of commercially available synthesizers. Only sentence-based experiments were used in this instance. The results are shown in Figs 16 and 17.

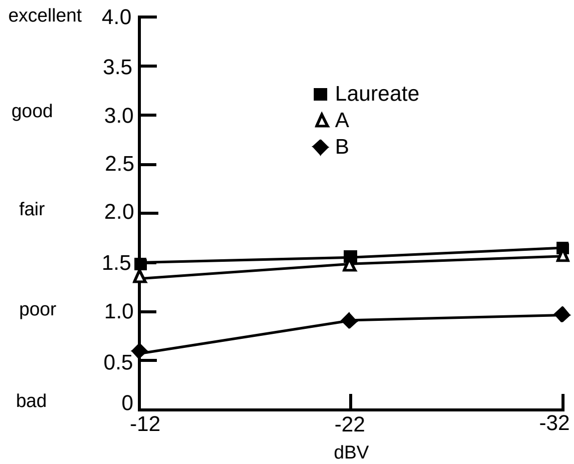

Fig 10 Experiment C — quality scale, NME sentences.

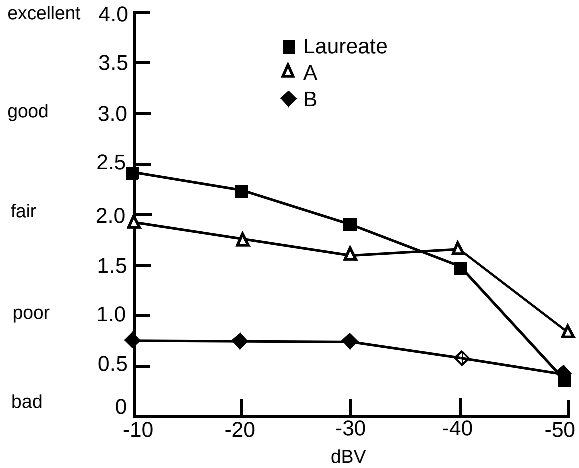

Fig 11 Experiment D — quality scale, paragraphs.

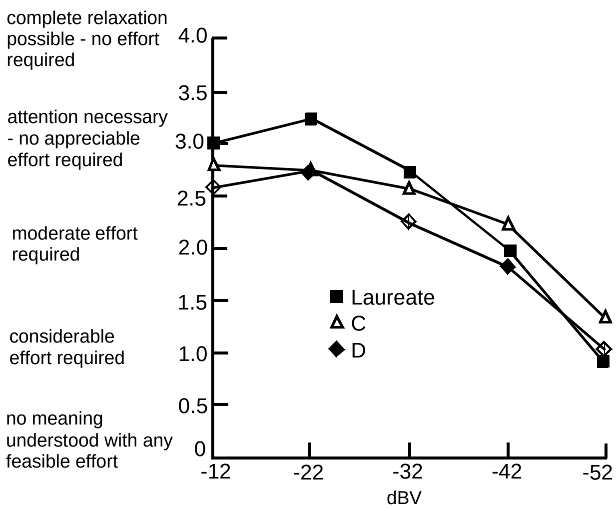

Fig 12 Experiment E — listening effort, NME sentences.

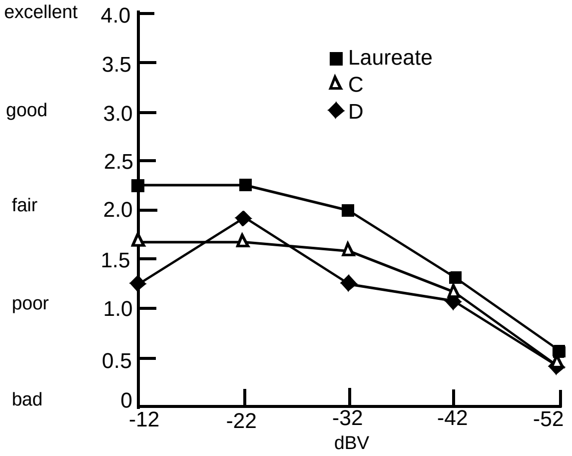

Fig 13 Experiment F — listening effort, paragraphs.

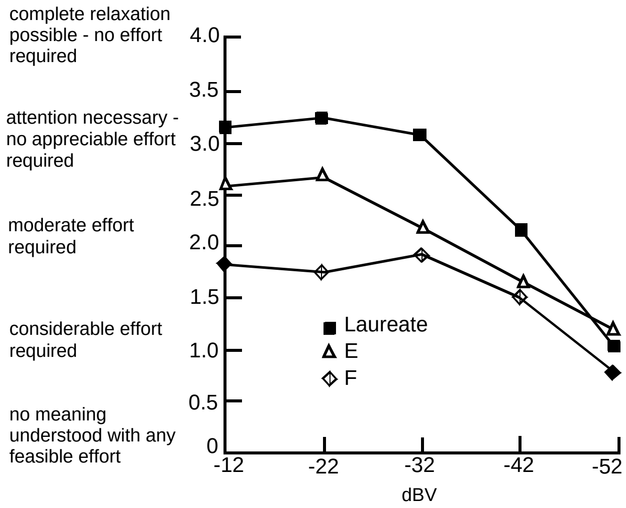

Fig 16 Experiment I — listening effort, NME sentences.

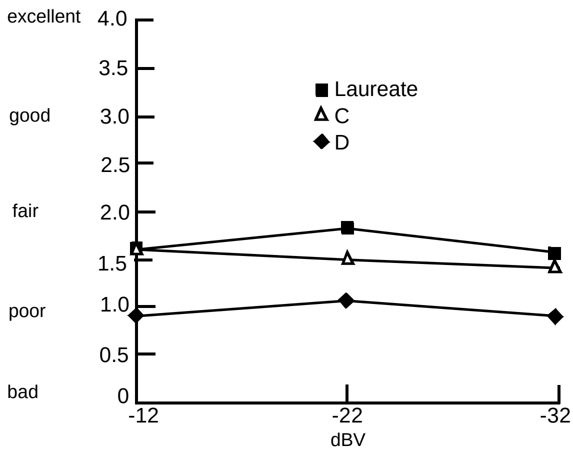

Fig 14 Experiment G — quality scale, NME sentences.

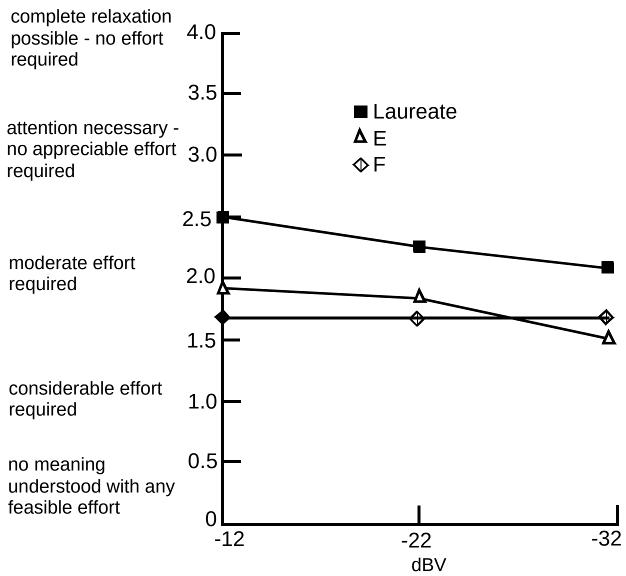

Fig 17 Experiment J — listening effort, paragraphs.

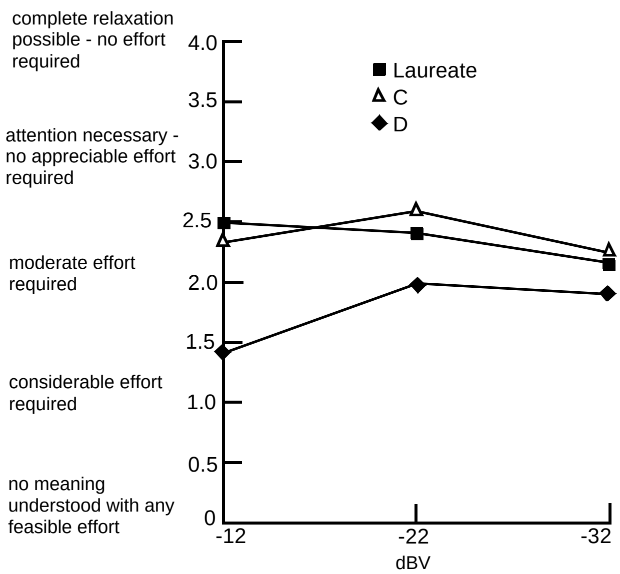

Fig 15 Experiment H — quality scale, paragraphs.

7.3 Robustness of the reference system

In Fig 18 all the 'reference equivalents' calculated for the one synthesizer (Laureate) which was common to all experiments is shown. There is a good correspondence for this in all the experiments with similar reference equivalents being obtained no matter what material or test method was used.

7.4 Discussion

The graphs of Figs 8-17 show the importance of level, as would be expected, and it is also clear that for most realistic listening levels the speech is regarded as being intelligible (i.e. it is scored above the category 'No meaning understood with any feasible effort') for every synthesizer when judged on the listening effort scale. It is also apparent, even from a casual examination of the orderings of preferences, that subjects rank the synthesizers in a very similar way, no matter what material or opinion is used.

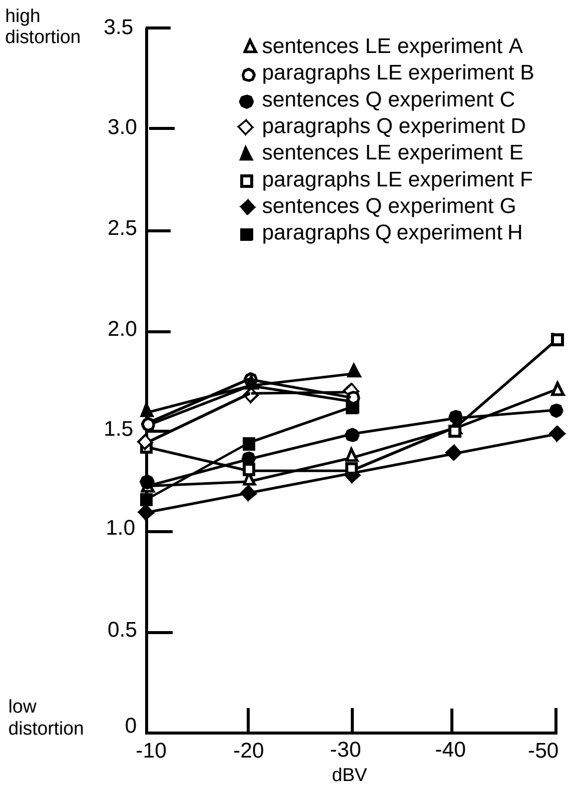

Fig 18 Reference TFW equivalent for Laureate found in eight different experiments.

As would be expected, the use of a different opinion scale produces a different set of absolute scores in all cases. Also the opinion scores recorded on paragraph material are lower (i.e. subjects are more critical) than is the case with sentence material. Figure 18 shows the extent to which these are normalized out by the reference system.

The listening effort scale appears to be more sensitive than the quality scale for this type of test, as measured by the range of categories used by subjects, and so is to be preferred for future tests. Also, despite the fact that the paragraph material is longer and elicited more critical responses it did not appear to provide better sensitivity than the original NME material.

7.5 Possible refinements and future developments

There are a number of areas where further investigation is required. In particular, it has not yet been established if the reference system is robust to different languages (a test in French is planned), and so far the system has only been used for 'male' voices — reflecting the fact that until recently there were very few 'female' voice text-to-speech synthesizers available. The reference system still lends itself to a number of refinements, e.g. it may be possible to find a distortion which is better at modelling the real distortions of TTS. The system is now being considered as a possible reference method for certain types of low-bit digital transmission system where the 'standard' MNRU distortion is inappropriate. It has also been considered for future multimedia assessment, particularly where it may be used to measure the perceived quality enhancement provided by having an associated picture of the speaker on a screen.

8. Conclusions

This chapter has described how a simple reference system based on time and frequency warping has been developed and how it has been used to provide a standard way for rating contemporary TTS systems. The system has been successfully used to provide a technology-independent means for monitoring in-house technology and for evaluating systems available in the market-place. It has been proposed as an ITU standard for evaluating synthetic speech systems.

References

1 Page J H and Breen A P: 'The Laureate text-to-speech system — architecture and applications', BT Technol J, 14, No 1, pp 57—67 (January 1996).

2 Edington M et al: 'Overview of current text-to-speech techniques: Parts I and II', BT Technol J, 14, No 1, pp 68—99 (January 1996).

3 Richards D L: 'Telecommunications by Speech', Butterworths (1973).

4 Hollier M P and Cosier G: 'Assessing human perception', BT Technol J, 14, No 1, pp 206—215 (January 1996).

5 Barrett P A et al: 'Speech transmission over digital mobile radio channels', BT Technol J, 14, No 1, pp 45—56 (January 1996).

6 CCITT Blue Book (1988).

7 Yaschin A et al: ‘Performance of speech recognition devices — evaluating speech produced over the telephone network’, ICASSP (1989).

8 Speigel M et al: ‘Using a monosyllabic test corpus to evaluate the intelligibility of synthesized and natural speech’, Proc of the American Voice I/O Systems Conf, San Franciso, USA (October 1988).

9 Voiers W D: ‘Evaluating processed speech using the Diagnostic Rhyme Test’, Speech Technology, 55, pp 30-39 (1983).

10 Cochran W G and Cox G M: ‘Experimental design’, Wiley (1957).

11 ITU: ‘Methods for the subjective assessment of small impairments in audio systems including multichannel sound systems’, Recommendation BS116 (November 1993).

12 SFERT: Systeme Fondamental European de Reference pour la Transmission Telephonique.
ARAEN: Appareil de Reference pour la Determination de l’Affaiblissement Equivalent pour la Nettete.
SRAEN: Systeme de Reference pour la Determination de l’Affaiblissement Equivalent pour la Nettete.
NOSFER: Nouveau Systeme Fondamental pour la Determination de l’Equivalent de Reference
[See CCITT White Book, Vol V (1968)].

13 CCITT Blue Book, Vol V: ‘Modulated Noise Reference Unit, P. 81’, (1988).

14 CCITT Blue Book, Vol V: ‘Methods used for assessing telephony transmission performance’, (1988).

15 Markel J D and Gray A H: ‘Linear prediction of speech’, Springer-Verlag (1976).

16 CCITT Blue Book , Vol V: ‘Sound rooms, P.48’, (1988)

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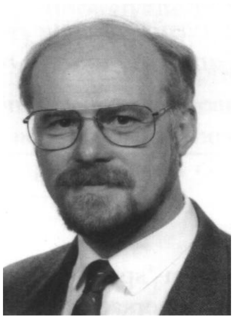

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